

GRADESENSE

- **Explainable AI for Paper Grade**

Transitions

- **Problem: predict off-spec basis-weight risk before limits are exceeded**
- **Team: GradeSense | Bhavuk Mahajan**
- **Honeywell Hackathon Final**

PROBLEM

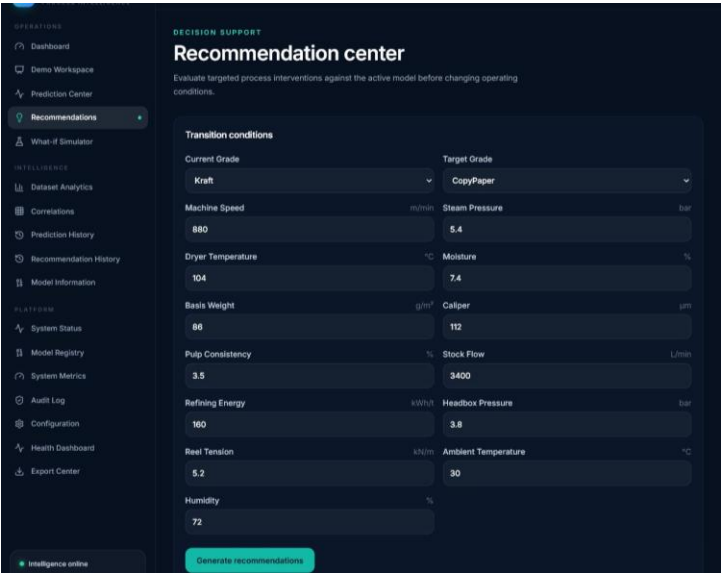
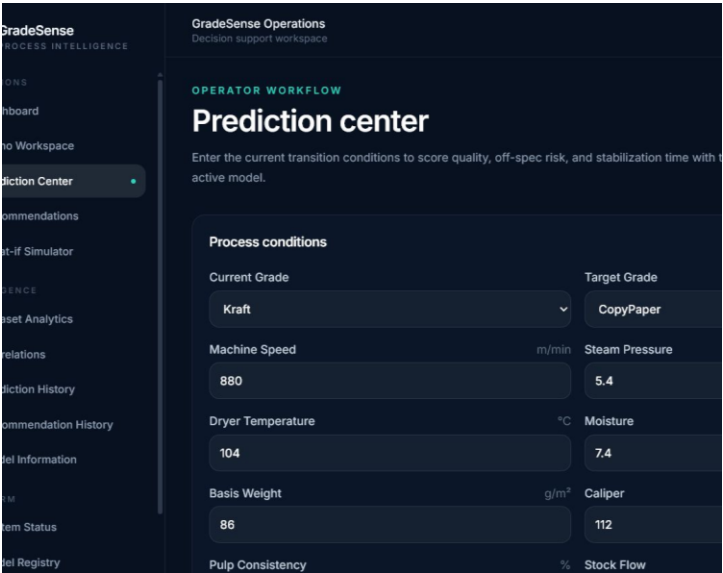
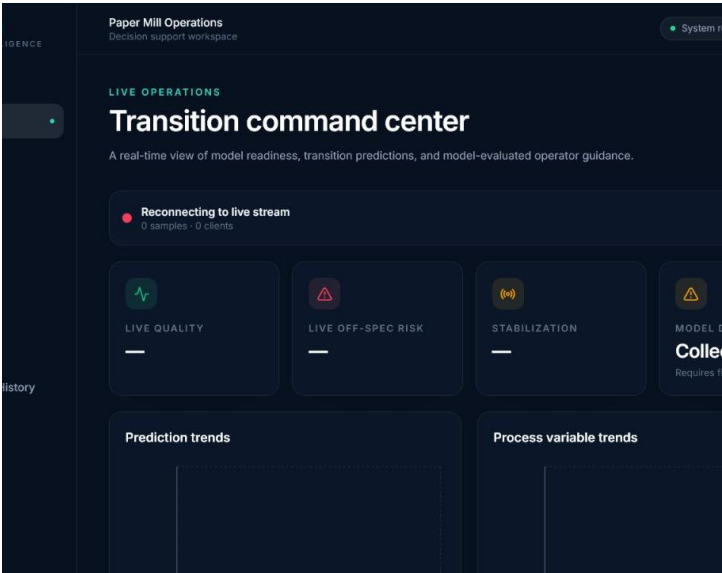
Paper grade transitions are difficult to stabilize because quality depends on interacting variables over time.

- Off-spec behavior is often detected after material has already been produced.
- Sensor trends, confidence, alerts, and historical outcomes are fragmented.
- Black-box predictions do not explain why risk changed or which action is safe.
- GradeSense must predict early, explain clearly, recommend safely, and keep operators in control.

TECHNICAL APPROACH

- React + TypeScript dashboard -> FastAPI REST and WebSocket layer -> domain services
- Snapshot ML + direct-horizon forecasting -> explainability -> safe recommendation ranking
- SQLAlchemy + PostgreSQL/SQLite persist predictions, alerts, decisions, outcomes, audit, and models
- Dockerized frontend/backend deployment on Render with health checks and environment configuration

SOLUTION DEMONSTRATION



INNOVATION + RESULTS

- Early basis-weight risk visibility supports intervention before avoidable off-spec production.
 - Direct-horizon forecasting reduces recursive error accumulation during grade transitions.
 - Explainable, constraint-checked recommendations keep operators in the decision loop.
 - Predictions, decisions, and outcomes form an auditable learning loop, deployed live on Render.

REFERENCES

- GitHub Repository
- <https://github.com/bhavukm007/GradeSense>
- Live Deployment
- <https://gradesense-4weh.onrender.com/>
- Demo Video
- <https://drive.google.com/file/d/1rCr3CzuCit0O029VzyxIhHoR1lv9E93E/view?usp=sharing>